Hirelytics – Project Planning, Literature Review & Requirement Gathering

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Group Code: ALX5_DAT3_G1

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1. Project Planning and Management

Hirelytics will be delivered before the final DEPI deadline of 4 Dec 2026 by a team of 6 members, split into a data layer (SQL, Excel), an analysis layer (Python) and a reporting layer (Power BI), under one team leader who owns the plan, the weekly checkpoints and the final submission.

1.1 Project Proposal: Overview

Call centers hire in high volume and lose many new hires early, yet hiring data and post-hire performance data usually sit in separate files. Hirelytics connects the two: it tracks every stage of the hiring process (applications to hire) and then follows the hired employees' performance KPIs, so the company can see which hiring sources, stages and candidate profiles produce the best agents.

1.2 Project Proposal: Objectives

- Build a clean, linked dataset from the two existing Excel sheets (hiring process and KPIs) by 12 Nov 2026.
- Analyze the hiring funnel: volume, conversion between stages, time to hire, and drop-off points.
- Analyze hired-employee performance against the agreed KPIs and identify top and bottom performers and trends.
- Link both phases to answer: which hiring factors (source, assessment score, interviewer, training result) predict performance and retention.
- Deliver an interactive Power BI dashboard with hiring KPIs and performance KPIs, plus a final report and presentation by 4 Dec 2026.

1.3 Project Proposal: Scope

In scope	Out of scope
Data cleaning, modeling and validation of the 2 Excel sheets	Collecting new data through surveys or live systems
Hiring-process analysis (funnel, time to hire, source effectiveness)	Building a real-time or automated data pipeline
Performance analysis of hired employees against KPIs	Payroll, compensation and non-call-center roles
Linking hiring attributes to performance (EDA and statistics in Python)	Deploying a production machine learning system
Power BI dashboard, final report and presentation	Making hiring decisions for the company

1.4 Task Assignment & Roles

Role	Main responsibilities	Main tools
Team Leader / Project Manager	Plan, task tracking, weekly meetings, risk log, stakeholder and mentor communication, final submission	WhatsApp
Data Engineer	Data cleaning, standardization, SQL database design (tables, keys, relationships), data validation	Excel, SQL

Hiring Analyst	Hiring funnel, time to hire, source effectiveness, hiring KPIs	SQL, Excel, Power BI
Performance Analyst	Employee performance KPIs, trends, ranking, retention by cohort	SQL, Excel, Power BI
Data Scientist	Exploratory analysis, statistical tests, correlation between hiring factors and performance	Python (pandas, matplotlib, seaborn, scipy)
Power BI Developer	Data model, DAX measures, dashboard design and publishing	Power BI
Documentation and QA Lead	Literature review, requirements document, data dictionary, report, presentation, quality checks	Word or Docs, PowerPoint

1.5 Resource Allocation: Tools

Tool	Used for
Excel	First-pass cleaning, pivot tables for quick checks, data dictionary, KPI definitions
SQL	Building the relational model, joining hiring data to performance data, KPI queries
Python	Exploratory data analysis, statistical tests, visual exploration, optional predictive model for performance
Power BI	Final data model, DAX measures, interactive dashboard (hiring page, performance page, linkage page)

GitHub	Version control for SQL scripts and notebooks, shared project files
Google Meet	Task board, weekly deadlines, meeting notes

1.6 Project Plan: Timeline, Milestones & Deliverables

Milestone	DEPI deadline	Deliverables
Planning, literature review and requirements	Oct 16, 2026	This document, uploaded to GitHub
System analysis and design	Nov 6, 2026	Problem statement, use cases, ER diagram, logical and physical schema, data flow diagram, dashboard wireframes, technology stack
Implementation	Nov 30, 2026	Cleaned data and SQL model, Python notebooks, Power BI dashboard, GitHub repository with README
Final presentation, testing and reports	Dec 4, 2026	Test plan and results, user manual, technical documentation, presentation, project infographic

1.7 Management approach

- **Meetings:** one 30-minute stand-up twice a week and one 60-minute review at the end of each week to demo progress against that week's milestone.
- **Task tracking:** every task has one owner, one due date and a status (To do, In progress, Review, Done) on a shared board.
- **Quality control:** every SQL query, DAX measure and chart is checked by a second member before it enters the report.

- **Version control:** files carry a version and date in the name; the data model and notebooks have one master copy. All project documents are uploaded to GitHub, as the DEPI guidelines require.

1.8 Risk Assessment & Mitigation Plan

Risk	Likelihood	Impact	Mitigation
Data quality issues (missing values, inconsistent IDs between the two sheets)	High	High	Start cleaning on 24 Oct, agree on one employee ID rule, log every cleaning decision
The two sheets cannot be linked reliably	Medium	High	Test the join on a sample before 6 Nov; fall back to cohort-level comparison if row-level linking fails
Uneven skills across the team	Medium	Medium	Pair a stronger and a newer member on each tool; share short how-to notes
Scope creep (extra analyses and pages)	Medium	Medium	Freeze scope on 16 Oct; new ideas go to a Phase 2 list
Member unavailable near the deadline	Medium	High	Every role has a named backup; finish the dashboard by 30 Nov, not in the final week
Sensitive employee data	Medium	High	Anonymize names, use IDs only, and keep the files in the shared team space

1.9 Project KPIs

These proposed targets measure whether the project itself succeeds; the team should confirm them at the first stand-up.

KPI	Proposed target	How it is measured
On-time delivery	4 of 4 DEPI deadlines met	Submission dates on GitHub
Task completion	90% or more of planned tasks done at each milestone	Task board status
Data quality	95% or more of hired candidates matched to a KPI record, no duplicate IDs in the final model	SQL validation queries
KPI accuracy	100% of dashboard KPIs match a manual Excel check	Second-member review log
Dashboard response time	Each page loads and filters in 3 seconds or less	Timed test in Power BI before 30 Nov
Stakeholder feedback	80% or more of reviewers rate the dashboard clear and useful	Feedback form after the demo

2. Literature Review

Published work shows that pre-hire data and early performance data can predict who stays and who performs in call centers, and that call center KPIs are well standardized. In the sources reviewed here, hiring analytics and performance KPIs are mostly studied separately, which is the gap Hirelytics fills by analyzing both phases together.

2.1 Hiring data and turnover in call centers

- **Pre-hire predictors.** A City University of Hong Kong doctoral study of 442 agents at a multinational logistics company found that biographical data from the application, such as the number of previous employers in the last five years and having reapplied to the company, were significant predictors of voluntary turnover (CityU thesis).
- **Using HR system data.** A SHRM case on a financial services firm built models on five years of historical personnel data to estimate how likely call center employees were to quit within a month and within a year (SHRM).

- **Performance data improves prediction.** A study in Applied Artificial Intelligence used 2,407 sales agents' operational records with Random Forest and naive Bayes classifiers, and found that adding performance changes over time raised turnover prediction accuracy up to 85% (Applied Artificial Intelligence, 2015). This supports linking the hiring and performance phases.
- **Cost of early exits.** A Korn Ferry article reports, citing other studies, that 46% of 20,000 new call center hires left within 18 months and that turnover costs about three to four months of pay per employee (Korn Ferry). The team should check the original studies before quoting these numbers.

2.2 Call center performance KPIs

Industry guides agree on a core set of agent KPIs. Benchmarks vary by call center type, so Hirelytics will compare agents against the company's own targets and averages, not generic numbers.

KPI	What it measures	Note from the literature
Average Handle Time (AHT)	Talk time + hold time + after-call work, per interaction	Should be read together with FCR and CSAT, because pushing AHT down alone can hurt resolution
First Call Resolution (FCR)	Share of issues resolved on the first contact	Core quality KPI
Quality Assurance (QA) score	Weighted score from call evaluations (accuracy, handling, compliance)	Counterweight to efficiency KPIs
Schedule adherence	Time in adherence ÷ total scheduled time × 100	A commonly cited healthy level is 90% or above
Customer satisfaction (CSAT)	Customer rating after the interaction	Pairs with QA and FCR

Time to proficiency	Time for a new hire to reach target performance	A vendor guide cites 60 to 90 days for most roles; useful to link hiring to performance

Sources: HiveDesk, Zendesk, COPC scorecard guide, Zizo reference.

2.3 Hiring process KPIs

These are standard HR recruitment metrics, which the team will confirm against the columns of the hiring sheet:

- Applicants per stage and stage-to-stage conversion rate (application, screening, interview, assessment, offer, hire).
- Time to hire: days from application to offer acceptance or start date.
- Source of hire: which channel (referral, job board, walk-in, social media) brings the most hires and the best performers.
- Offer acceptance rate and no-show rate before training starts.
- Early attrition: share of hires who leave within the first 30, 60 or 90 days.
- Quality of hire: the performance KPIs of hired employees, which is the link Hirelytics builds.

2.4 Methods and tools in similar projects

Similar HR analytics work follows the same order: clean and join the data (Excel, SQL), explore and test relationships between candidate attributes and outcomes (Python statistics, and classification models when the data is large enough), then communicate results through a dashboard (Power BI). Hirelytics follows this order. A predictive model is optional and will only be built if the cleaned data has enough rows and a clear outcome column.

2.5 Gap and contribution

- Reviewed studies focus mainly on predicting turnover, not on following the whole funnel from application to KPI performance.
- Industry KPI guides cover performance KPIs but not how hiring decisions shape them.
- Hirelytics contributes one connected view: hiring funnel KPIs, employee performance KPIs, and the factors linking them, in a dashboard a recruiter or operations manager can use.

2.6 Feedback & Evaluation

To be completed by the lecturer after reviewing this document.

2.7 Suggested Improvements

To be completed by the lecturer. The team will record each suggestion here and mark it as done or planned.

2.8 Final Grading Criteria

The DEPI guidelines base the final marks on documentation, implementation, testing and presentation. The mark breakdown will be added here once the lecturer provides it.

3. Requirement Gathering

Hirelytics has two data sources already in hand, an Excel sheet for the hiring process and an Excel sheet for the KPIs, so the requirements below define what the team must build from them and what must be confirmed before 6 Nov. Field names are expected fields and must be checked against the real sheets.

3.1 Stakeholder Analysis

Stakeholder	Interest in Hirelytics
Recruitment / HR team	Which sources, stages and profiles give the best hires, and where candidates drop out
Operations managers and team leaders	How new hires perform against KPIs and how fast they reach target
Trainers and QA team	Which hire groups need more training or coaching
Senior management	Hiring cost and quality, and early attrition
Project team and DEPI mentor	A complete, accurate and well-documented graduation project

3.2 User Stories & Use Cases

ID	User story
US1	As a recruitment manager, I want to see how many candidates reach each hiring stage, so that I can find where we lose them.
US2	As a recruiter, I want to compare hiring sources by hires and later performance, so that I can focus on the best channels.
US3	As an operations manager, I want to see each team's KPIs against target by month, so that I can spot teams that need support.
US4	As a trainer or QA lead, I want to see how new hires perform in their first months, so that I can plan coaching.
US5	As senior management, I want to see early attrition by hiring batch and source, so that I can reduce hiring cost.
US6	As a project analyst, I want one validated and linked dataset, so that every KPI comes from the same source.

Use case	Actors	Flow	Result
UC1 View hiring funnel	Recruiter, HR manager	Opens the hiring page, filters by period and source	Stage conversion, drop-off and time to hire
UC2 Review performance	Operations manager, team leader	Opens the performance page, filters by team and month	KPIs against target, top and bottom teams or agents

UC3 Analyze hiring to performance link	HR analyst, QA lead	Opens the linkage page, picks a source or assessment band	Performance and early attrition by hiring profile
UC4 Refresh and validate data	Project data engineer	Loads updated sheets, runs the validation queries	Clean model with row counts that match the source

3.3 Business questions

Hiring phase

- How many candidates enter each stage, and what share moves to the next stage?
- How long does hiring take, and where is the biggest delay?
- Which sources and recruiters bring the most hires?

Performance phase

- How do hired employees perform on each KPI over their first months?
- Which agents, teams or hiring batches are above or below target?
- How many hires leave early, and when?

Linking both phases

- Do assessment scores, interview results or hiring source predict later performance?
- Which hiring profile gives the best performance and retention?

3.4 Data requirements

Source	Expected content (to confirm)	Used for
Hiring process sheet	Candidate ID, application date, source, recruiter or interviewer, status and date for each stage, assessment scores, offer and hire dates, rejection reason	Hiring funnel, time to hire, source effectiveness
KPI sheet	Employee ID, period (day, week or month), team, AHT, FCR, QA score, schedule adherence, CSAT, attendance, status and exit date	Performance analysis, early attrition

Link between sheets	One shared ID (candidate ID = employee ID) or a mapping table	Linking hiring factors to performance

Data checks to run before the 6 Nov design deadline:

1. Row counts, date range and number of unique candidates and employees in each sheet.
2. Share of hired candidates that can be matched to a KPI record.
3. Missing values, duplicate IDs, mixed date formats and inconsistent category names (for example, source spelled in several ways).
4. Outliers in KPI values, such as impossible AHT or scores above 100%.

3.5 KPI requirements

Phase	KPI	Definition
Hiring	Stage conversion rate	Candidates reaching a stage ÷ candidates in the previous stage
Hiring	Time to hire	Days from application date to hire date
Hiring	Source effectiveness	Hires ÷ applicants, per source
Hiring	Offer acceptance rate	Offers accepted ÷ offers made
Hiring	Early attrition rate	Hires who leave within 90 days ÷ total hires
Hiring	Cost per hire	Total hiring cost ÷ hires (only if cost data exists)
Performance	Average Handle Time	(Talk + hold + after-call work) ÷ number of calls

Performance	First Call Resolution	Issues resolved on first contact ÷ total issues
Performance	QA score	Average quality evaluation score
Performance	Schedule adherence	Time in adherence ÷ scheduled time × 100
Performance	CSAT	Average customer satisfaction rating
Performance	Time to proficiency	Days until a new hire first reaches the target KPIs
Linkage	Quality of hire	Share of hires meeting target KPIs after 90 days, by source or profile

The team will confirm which of these the two sheets can support and drop or rename those that cannot be calculated.

3.6 Functional Requirements

ID	Requirement	Priority
FR1	Clean and standardize both sheets and document every cleaning step	Must
FR2	Build a linked data model with an employee/candidate table, a date table and the two fact tables	Must
FR3	Calculate all agreed hiring and performance KPIs with documented formulas	Must

FR4	Show the hiring funnel with conversion and drop-off at each stage	Must
FR5	Show performance KPIs by agent, team, month and hiring batch against targets	Must
FR6	Show how hiring source and assessment results relate to performance and early attrition	Must
FR7	Provide filters for date, source, recruiter, team and status	Should
FR8	Provide a short list of findings and recommendations for HR and operations	Should
FR9	Build a predictive model for early attrition or low performance	Could

3.7 Dashboard requirements

- **Page 1, Hiring overview:** funnel, time to hire, source comparison, hires per month.
- **Page 2, Performance overview:** KPI cards against target, trend by month, top and bottom teams or agents.
- **Page 3, Hiring to performance:** performance by source and by assessment score band, early attrition by hiring batch.
- Common to all pages: the same filters, consistent colors, KPI cards with target comparison, and a one-line explanation of each KPI.

3.8 Non-functional Requirements

- **Accuracy:** every KPI is checked by a second team member against a manual calculation in Excel.
- **Privacy:** names and contact details are removed or replaced with IDs before sharing.
- **Usability:** a non-technical manager can read each page in under two minutes.
- **Reproducibility:** SQL scripts, Python notebooks and the Power BI file are stored together with a data dictionary.
- **Performance:** the dashboard refreshes and filters without noticeable delay on the project dataset.

3.9 Assumptions and constraints

- The two Excel sheets are the only data sources and are final for this project.
- Both sheets can be linked by a shared ID, or a mapping can be built before 6 Nov.
- The project runs from 14 Oct to 4 Dec 2026 and team members work part-time next to their other DEPI tasks.
- Targets for each KPI come from the company's own standards if the sheets include them, and from the average of the data if not.

3.10 Deliverables and acceptance criteria

Deliverable	Accepted when
Cleaned dataset and SQL model	Row counts reconcile with the original sheets and the data dictionary is complete
Python analysis notebook	It runs from start to end and every chart answers a listed business question
Power BI dashboard (3 pages)	All Must requirements are met and the KPI values match the Excel checks
Final report	It covers planning, literature review, requirements, method, findings and recommendations
Final presentation	The team can present it in the allotted time and answer questions on any section